Steven Koehl

AEROSPACE ENGINEERING | TECHNICAL OPERATIONS | SOFTWARE DEVELOPMENT

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PROFESSIONAL SUMMARY

U.S. Air Force Technical Sergeant with 18 years of experience across aerospace maintenance, avionics diagnostics and circuit-level repair, technical program coordination, product quality reporting, contract oversight, and technical instruction. Holds a Master of Science in Aeronautical Engineering from Embry-Riddle Aeronautical University. Combines hands-on engineering experience and operational leadership with independent software development, AI integration, and workflow automation.

CORE CAPABILITIES

Aerospace & Engineering
Avionics systems; circuit-level diagnostics; electronics repair; maintenance and reliability; technical documentation

Technical Operations
Program coordination; product quality reporting; contract oversight; process improvement; workforce instruction

Software Development
Python; TypeScript; JavaScript; React; Next.js; C++; web applications and PWAs

Infrastructure & Emerging Technology
Linux; Docker; networking; cloud deployment; self-hosted systems; AI integration; OCR and automation

PROFESSIONAL EXPERIENCE

Senior AFREP Technician & AI Developmental Lead

United States Air Force | Fairchild Air Force Base | Current

- Serve as a senior technician in the Air Force Repair Enhancement Program, supporting equipment sustainment and technical problem-solving.
- Lead evaluation and practical application of AI tools for aerospace maintenance workflows.
- Coordinate technical programs involving Technical Order distribution, product improvement, and service-contract oversight.

Product Quality Deficiency Reporting Liaison

United States Air Force | Liaison to Air Force Materiel Command | Prior role

- Served as the primary liaison to Air Force Materiel Command for the Joint Deficiency Reporting System (JDRS).
- Coordinated deficiency investigation, technical documentation, reporting, and follow-up with engineering and maintenance stakeholders.
- Supported corrective-action tracking and communication of product quality issues affecting aircraft systems.

Technical Program Coordination

United States Air Force | Prior role

- Coordinated Technical Order distribution, product improvement activities, and service-contract oversight across maintenance programs.
- Monitored technical information workflows and contract performance requirements.
- Translated maintenance and equipment issues into documented actions for program and engineering stakeholders.

Technical Instructor - Electronics & AFREP Curriculum

United States Air Force | Sheppard Air Force Base | Prior role

- Developed and delivered technical instruction in complex electronics, micro-miniature soldering, and component-level diagnostics.
- Applied Instructional System Design principles to structure technical lessons, practical exercises, and assessment.
- Coached technicians in diagnostic methods, repair practices, and safe laboratory procedures.

Avionics Technician - Circuit-Level Maintenance

United States Air Force | 2008-Present

- Performed circuit-level troubleshooting, diagnostics, and repair on aircraft avionics and electronic systems.
- Applied micro-miniature circuit-card repair, soldering, and component analysis techniques.
- Supported aerospace maintenance operations through technical problem-solving and equipment sustainment.

SELECTED SOFTWARE & ENGINEERING PROJECTS

Clivus | C++20, Qt 6, DCMTK, OpenGL
Open-source Windows DICOM viewer focused on image fidelity, responsive interaction, and test-backed medical imaging workflows.

VerbaLibera | Next.js, TypeScript, PWA
Offline-capable language-learning application built around learner-driven sentence construction.

OpenNarrator | Python, FastAPI, Typer
Open-source local audiobook generation tool with web, command-line, and automation interfaces.

92 AMXS Tracker | React, TypeScript, Firebase, OCR
Maintenance workflow application with role-based access, offline support, and document intake automation.

EDUCATION & PROFESSIONAL DEVELOPMENT

Master of Science in Aeronautical Engineering
Embry-Riddle Aeronautical University

Instructional System Design; Contracting Officer Representative oversight; Joint Deficiency Reporting System / AFMC liaison experience; AI integration and workflow automation initiatives.

Additional details and project source code: stevenkoehl.com | github.com/sleuthy-sloth